

8.8 Snub Cube and Snub Dodecahedron: Cardinality of Chirality

1. Why These Two Shapes Matter

The snub cube and the snub dodecahedron are not just examples of snub truncation.

They are the clearest geometric witnesses of chirality inside the Archimedean/Catalan bridge.

In OMI, that matters because the model is not only trying to describe form. It is trying to describe how two addressed views can share a center, differ by handedness, and still preserve a lawful transformation path.

The snub cube and snub dodecahedron show this directly.

They are:

```
highly structured  
highly symmetric  
not mirror-symmetric  
left/right handed  
triangle-rich  
generated by cutting, twisting, and re-facing
```

This is why they are better OMI examples than simple permutation or circular inversion.

Permutation reorders.

Circular inversion reverses.

Snub truncation transforms.

It cuts a parent body, twists the relation, and fills the boundary with triangular faces.

That is the OMI operation.

```
The hyphen joins.  
The snub twists.  
The triangle agrees.  
The centroid holds.
```

2. Snub Truncation as the OMI Operation

In classical geometry, snub truncation begins with a parent polyhedron.

The operation does not merely slice the corners. It separates the original faces, rotates them, and fills the gaps with new equilateral triangles.

The operation has three visible stages:

```
cut      = remove vertex excess
twist    = introduce handed orientation
re-face  = create new triangular boundary faces
```

In OMI terms:

```
cut      = isolate unresolved boundary pressure
twist    = pass the relation through the hyphen bridge
re-face  = surround the transformed relation with agreement faces
record   = preserve the accepted transformation path
project  = expose the new state without erasing the old one
```

So the OMI operation is not overwrite.

```
OMI snubs; it does not overwrite.
```

An object does not simply become another object. It is transformed through a bounded operation. The previous body remains replayable. The new body becomes an accepted configuration.

3. The Shared Centroid

The most important OMI idea is that these transformations can be understood as happening around a shared centroid.

If the Platonic solids, Archimedean solids, Catalan duals, and 4D regular polytope projections are rooted around the same tetrahedral centroid, then the snub cube and snub dodecahedron expose something special:

```
they show the cardinality of chirality
```

They show that the same centered structure can resolve into two non-superimposable orientations:

left-handed
right-handed

The centroid remains shared.

The orientation differs.

That is the P2P insight.

In a peer-to-peer situation, two peers do not need to collapse into the same view. They need to share a centroid of reference and preserve a lawful transformation path between their handed states.

The snub solids show how this can happen geometrically.

same centroid
different handedness
lawful relation
replayable transformation

This is the geometric meaning of agreement without collapse.

4. Two's-Complement Analogy

The two chiral forms can be read like a geometric two's-complement pair.

They are not identical.

They are not unrelated.

They are complementary orientations around the same centered body.

In binary arithmetic, two's complement lets positive and negative values share the same bounded word space. In OMI geometry, left-handed and right-handed snub forms share the same bounded centroid space.

So the snub pair can be read as:

left snub = one orientation of the bounded body
right snub = complementary orientation of the bounded body

The important point is not that geometry is literally binary arithmetic. The point is that both systems encode opposition inside a bounded domain.

OMI uses this as a model for peer agreement:

```
peer A may hold one orientation
peer B may hold the complementary orientation
agreement occurs when both resolve around the same centroid
```

This is not mirror collapse.

It is centroid-preserving complementarity.

5. The Cube Path

Start with the regular cube.

The cube has:

```
6 square faces
8 vertices
12 edges
```

Standard Truncation of the Cube

In standard truncation, the 8 corners are sliced off.

Each removed corner becomes a new triangular face.

The original square faces expand into octagons.

The truncated cube has:

```
8 triangular faces
6 octagonal faces
14 total faces
```

Standard truncation is a cut operation.

It removes vertex excess, but it does not create the same kind of chiral twist.

Snub Operation of the Cube

In the snub operation, the original square faces are separated and rotated.

The gaps are packed with new equilateral triangles.

The snub cube has:

```
6 square faces
32 triangular faces
38 total faces
60 edges
24 vertices
```

At each vertex, one square and four triangles meet.

That is the key.

The cube begins as a simple Platonic parent.

The snub cube becomes a chiral Archimedean body.

The new triangular faces show how much boundary had to be re-faced to preserve the transformed relation.

In OMI terms:

```
cube parent      = stable local body
snub cube        = chiral transformed body
32 triangles     = agreement faces generated by the twist
left/right forms = cardinality of chirality
```

The snub cube is the cubic witness that the hyphen bridge does more than connect terms.

It twists them into orientation.

6. The Dodecahedron Path

Start with the regular dodecahedron.

The dodecahedron has:

12 pentagonal faces
20 vertices
30 edges

Standard Truncation of the Dodecahedron

In standard truncation, the 20 corners are sliced off.

Each removed corner becomes a new triangular face.

The original pentagons expand into decagons.

The truncated dodecahedron has:

20 triangular faces
12 decagonal faces
32 total faces

Again, standard truncation is mostly a cutting operation.

It changes the boundary, but it does not expose the same degree of chirality.

Snub Operation of the Dodecahedron

In the snub operation, the 12 pentagons are separated and rotated.

The gaps are filled with equilateral triangles.

The snub dodecahedron has:

12 pentagonal faces
80 triangular faces
92 total faces
150 edges
60 vertices

At each vertex, one pentagon and four triangles meet.

This is the richer version of the same chiral principle.

The dodecahedron begins as a Platonic parent.

The snub dodecahedron becomes a chiral Archimedean body.

The 80 triangles show the larger agreement surface needed to preserve the transformed relation.

In OMI terms:

```
dodecahedron parent = higher Platonic body
snub dodecahedron   = chiral transformed body
80 triangles         = expanded agreement faces
left/right forms    = larger chirality witness
```

The snub dodecahedron exposes chirality at a higher cardinality than the snub cube.

7. Cube vs. Dodecahedron

The two snub examples form a clean comparison.

Parent	Snub Solid	Original Faces Preserved	New Triangles	Total Faces	Vertices	Edges
Cube	Snub cube	6 squares	32	38	24	60
Dodecahedron	Snub dodecahedron	12 pentagons	80	92	60	150

The pattern is important.

Each snub solid keeps the parent face type:

```
cube keeps squares
dodecahedron keeps pentagons
```

But the snub operation surrounds those original faces with triangles.

The triangles are the new agreement faces created by the twist.

This gives OMI a way to talk about transformation:

```
the parent identity is preserved
the relation is twisted
the boundary is re-faced
the new state gains chirality
```

That is exactly what an OMI object should do when it changes without losing its address.

8. Archimedean/Catalan Bridge

The snub cube and snub dodecahedron belong to the Archimedean family.

Their duals belong to the Catalan family.

This is why the OMI `16xy` term belongs to the Archimedean/Catalan bridge.

The bridge is where a parent body becomes transformed, dualized, twisted, and re-faced.

```
4y2   = Platonic vertex seed
16xy   = Archimedean/Catalan snub bridge
60x2  = 4D regular polytope orbit
```

The Platonic body gives the seed.

The Archimedean/Catalan bridge performs the transformation.

The 4D regular polytope orbit gives the larger closure field.

So the OMI quadratic form can be read geometrically:

$$Q_{xy}(x, y) = 60x^2 + 16xy + 4y^2$$

as:

```
cosmic orientation orbit
+
snub bridge
+
atomic Platonic seed
```

Or in the order of emergence:

```
4y2 seeds.
16xy snubs.
60x2 closes.
```

9. The Fano Lottery

The snub operation creates many triangular faces, but OMI does not treat those faces as random.

The Fano layer gives a finite incidence selector.

A Fano plane has:

```
7 points
7 lines
3 points per line
```

OMI reads the truth row as a triangular incidence unit:

```
(LL, NN, MM)
```

That truth row can be read as:

```
lens
antecedent
consequent
```

or:

```
declaration
definition
receipt
```

or:

```
source
frame
projection
```

The “Fano lottery” is therefore not chance. It is bounded selection from finite incidence.

It asks:

```
Which triangle resolves?
Which line intersects?
```

Which relation closes?
Which receipt can be written?

The snub bridge produces the triangular agreement field.

The Fano layer selects the resolving triangle.

The replay ring remembers the accepted path.

10. Cardinality of Chirality

The snub cube and snub dodecahedron expose chirality in two related ways.

First, each has two handed forms:

left-handed
right-handed

That gives the minimal cardinality of chirality:

chirality cardinality = 2

Second, each shape expresses that handedness through a different scale of triangular agreement faces:

snub cube = 32 triangles
snub dodecahedron = 80 triangles

So the chirality is binary in handedness, but scaled in surface complexity.

OMI can read this as:

2 orientations
many agreement faces
one shared centroid

That is the important relation.

A peer-to-peer system does not need every peer to see the same handed surface. It needs the peers to share the centroid and preserve the transformation path.

Thus:

```
chirality is not disagreement  
chirality is oriented complementarity
```

Agreement is the ability to resolve the relation through the shared centroid.

11. Why Snubbing Is Better Than Permutation or Inversion

Permutation is useful, but it is not enough.

Permutation says:

```
same elements  
different order
```

Circular inversion is useful, but it is also not enough.

Circular inversion says:

```
inside becomes outside  
outside becomes inside
```

Snub truncation says more:

```
the body is cut  
the relation is twisted  
the boundary is re-faced  
chirality appears  
the parent remains recoverable  
the new form becomes uniform
```

That is the OMI transformation.

OMI is not merely rearranging terms.

OMI is not merely turning a shape inside out.

OMI is joining terms through a hyphen field and transforming them into an oriented, replayable agreement surface.

This is why the snub operation belongs at the center of the OMI geometric explanation.

12. OMI Meta-Compiler Reading

The hyphen is the join operation.

In ordinary language, a hyphen joins terms into one compound word:

```
open-world  
source-object  
meta-compiler
```

In OMI, the hyphen joins address terms into one bounded instruction word:

```
omi---imo
```

The hyphen field is not empty space.

It is the active middle:

```
join field  
rewrite bus  
gravity field  
snub bridge  
compiler lane
```

So the snub operation is the geometry of the hyphen.

```
omi enters  
--- joins and twists  
imo closes
```

When written as a compiler path:

```
omi-ip4-gravity-ip6-imo
```

the same doctrine appears:

```
compact address enters  
hyphen field rewrites
```

expanded address resolves
closure records

The snub cube and snub dodecahedron are therefore not decorative examples. They are geometric explanations of what the hyphen field does.

The hyphen joins.
The snub transforms.
The triangle agrees.
The centroid holds.

13. Final Canon

The snub cube and snub dodecahedron are the two clearest Archimedean witnesses of OMI chirality.

They show that a body can share a centroid, preserve a parent identity, twist into left/right handedness, and gain a new triangular agreement surface without being overwritten.

The snub cube exposes this through:

6 squares
32 triangles
38 faces
2 handed forms

The snub dodecahedron exposes this through:

12 pentagons
80 triangles
92 faces
2 handed forms

Together, they show the cardinality of chirality:

two orientations
one centroid
many triangular agreement faces

OMI reads this through the quadratic bridge:

$4y^2$ = Platonic vertex seed
 $16xy$ = Archimedean/Catalan snub bridge
 $60x^2$ = 4D regular polytope orbit

Short doctrine:

Platonic seed.
Snub bridge.
Chiral pair.
Shared centroid.
Fano triangle.
Replay memory.

One-line canon:

The snub cube and snub dodecahedron expose OMI chirality: two handed orientations around one centroid, re-faced by triangular agreement and remembered as replayable transformation.